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Preparation of Solid Polyfunctional Alkynylzinc Pivalates with Enhanced Air and Moisture Stability for Organic Synthesis

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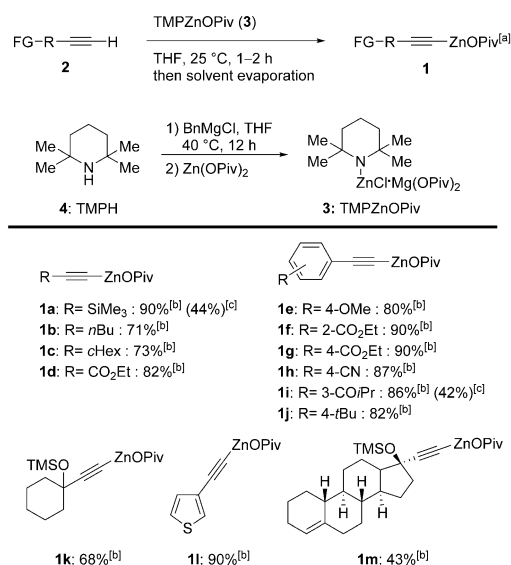
Abstract: We report the preparation of solid and air-stable polyfunctionalized alkynylzinc pivalates from the corresponding alkynes using TMPZnOPiv ($\text{TMP} = 2,2,6,6\text{-tetramethylpiperidyl}$) as base. These organozinc pivalates are obtained as powders under mild conditions in excellent yields and can be manipulated in air for several hours without significant decomposition. These zinc reagents show an excellent reactivity in various carbon–carbon bond-forming reactions and 1,3-dipolar cycloadditions. An alkynylzinc pivalate has been used to prepare a carboxyamidotriazole with potential antineoplastic activity in eight steps and 38 % overall yield.

Polyfunctional alkynes are important target molecules in material and medicinal chemistry.^[1] They are also key intermediates for the synthesis of other common functional groups such as *E*- and *Z*-alkenes.^[2] Alkynyl organometallics are privileged reagents for the synthesis of various functionalized alkynes.^[3] Recently, we have reported several methods for preparing organozinc reagents with enhanced air and moisture stability starting either from (hetero)aryl, allylic, and benzylic halides or from (hetero)arenes.^[4] These zinc organometallics were found to be valuable reagents for the high-throughput screening of biomolecules.^[5] Related solid Reformatsky enolates have been used for organic synthesis.^[6] (Hetero)arylzinc pivalates proved to be unique bench-stable solid reagents for cobalt-catalyzed cross-couplings.^[7] Herein, we describe the preparation of a new class of alkynylzinc pivalates of type **1** bearing various functional groups and displaying an enhanced air and moisture stability. They are obtained in high yields from the corresponding alkynes of type **2** by deprotonation using the mixed zinc–magnesium base $\text{TMPZnCl}\cdot\text{Mg}(\text{OPiv})_2$ (**3**; $\text{TMP} = 2,2,6,6\text{-tetramethylpiperidyl}$; abbreviated as TMPZnOPiv).^[8] Further, we have demonstrated the utility of alkynylzinc reagents (**1**) for the formation of new carbon–carbon bonds including their participation in 1,3-dipolar cycloadditions without the cleavage of the carbon–zinc bond. An application to the preparation of a carboxyamidotriazole with significant anticancer activity completes our study.

Preliminary studies have shown that the deprotonation of trimethylsilylacetylene (**2a**) with $\text{TMPZnOPiv}\cdot\text{LiCl}$ ^[4c] proceeded smoothly, but produced a solid alkynylzinc reagent

(**1a**) with moderate air stability (44 % activity after 4 h in air). We speculated that this problematic air and moisture sensitivity was due to the presence of LiCl and designed therefore a new synthesis of TMPZnOPiv (**3**) which did not contain LiCl . Thus, the treatment of TMP-H (**4**) with benzylmagnesium chloride in THF ^[9] (40 °C, 12 h) followed by the addition of $\text{Zn}(\text{OPiv})_2$ provided TMPZnOPiv (**3**) in quantitative yield. The deprotonation of **2a** with TMPZnOPiv (**3**, 1.2 equiv) afforded after solvent evaporation the corresponding alkynylzinc pivalate (**1a**) with highly improved air stability (90 % yield, determined after 4 h exposure to air) as shown in Scheme 1. This reactivity proved to be general and a range of polyfunctional alkynylzinc pivalates (**1a–1m**) were obtained with 43–90 % activity after 4 h of air exposure at 25 °C. Remarkably, a range of important functional groups such as a ketone, a nitrile, and an ester are tolerated in the zinc reagents of type **1**.

Furthermore, we examined the reactivity of these new solid alkynylzinc pivalates (**1**) in Negishi cross-coupling reactions^[10] with various aryl halides. The screening of several palladium catalytic systems showed that the Buchwald ligand, DavePhos,^[11] gave the best results in combination with $\text{Pd}(\text{dba})_2$ (2 mol %; $\text{dba} = \text{dibenzylideneacetone}$).^[12] Under these conditions, a range of (hetero)aryl iodides, bromides,



Scheme 1. Preparation of alkynylzinc pivalates of type **1** from the corresponding alkynes of type **2** using $\text{TMPZnCl}\cdot\text{Mg}(\text{OPiv})_2$ (abbreviated as TMPZnOPiv **3**). [a] Alkynylzinc chlorides and $\text{Mg}(\text{OPiv})_2$ complexes are abbreviated as alkynylzinc pivalates for clarity. [b] The indicated yields are obtained after 4 h of exposure to air. [c] Preparation with TMPZnOPiv in the presence of LiCl (1 equiv).

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and chlorides of type **5** underwent smooth Negishi cross-couplings producing arylated alkynes of type **6** (Table 1). Remarkably, a variety of sensitive functional groups were tolerated in these cross-couplings such as a ketone (entry 1 of Table 1), an aldehyde (entry 3), an unprotected indole (entry 4), and an amide (entry 7). The use of ethyl 6-chloronicotinate (**5b**) provided alkyne **6b** in 74 % yield

Table 1: Pd-catalyzed Negishi cross-coupling of solid alkynylzinc reagents of type **1** with various aryl halides leading to alkynes of type **6**.^[a]

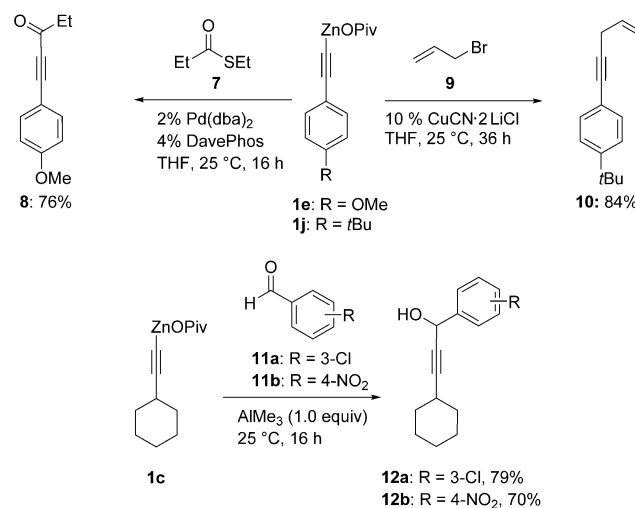
$\text{FG-R}-\text{C}\equiv\text{C}-\text{ZnOPiv} \xrightarrow[\text{THF, 25–60 } ^\circ\text{C}]{\text{Ar-X (5); X= Cl, Br, or I; Pd(dba)}_2 \text{ / DavePhos}} \text{FG-R}-\text{C}\equiv\text{C}-\text{Ar}$		
Entry	1	Product ^[b]
	Electrophile	
1	1a	5a 6a: 89 % (25 °C, 6 h)
2	1a	5b 6b: 74 % (60 °C, 16 h)
3	1b	5c 6c: 76 % (40 °C, 12 h)
4	1c	5d 6d: 68 % (50 °C, 2 d)
5	1e	5e 6e: 91 % (40 °C, 14 h)
6	1i	5f 6f: 93 % (40 °C, 16 h)
7	1i	5g 6g: 87 % (50 °C, 2 d)
8	1k	5h 6h: 80 % (40 °C, 12 h)
9	1l	5i 6i: 92 % (40 °C, 12 h)

[a] Pd(dba)₂ (2 mol %) and DavePhos (4 mol %) were used as catalyst.
[b] Yields refer to analytically pure products.

(entry 2), which was a key intermediate for the synthesis of tazarotene.^[13]

We also found that the alkynylzinc pivalates (**1**) undergo a smooth Pd-catalyzed acylation using thioesters as acylation reagents, a reaction pioneered by Fukuyama.^[14]

The use of 4 % DavePhos in combination with 2 % Pd(dba)₂ proved to be an excellent catalytic system. Thus, the alkynylzinc pivalate **1e** reacted with the thioester **7** providing alkynyl ketone **8** in 76 % yield (Scheme 2). Furthermore the



Scheme 2. Acylation, allylation, and aldehyde addition using solid alkynylzinc pivalates of type **1**.

alkynylzinc pivalate **1j** underwent a copper-catalyzed allylation with allyl bromide (**9**) using 10 % CuCN·2LiCl^[15] as catalyst to afford enyne **10** in 84 % yield. Although such alkynylzinc reagents did not react readily with ketones, a smooth addition to aldehydes was promoted by AlMe₃ as reported by Woodward.^[16] Thus, when alkynylzinc pivalate **1c** was added to benzaldehydes **11a,b** at 25 °C, propargylic alcohols **12a,b** were obtained within 16 h in 79 and 70 % yield, respectively.

In 2002, Sharpless, Fokin, and Meldal reported copper-catalyzed regioselective azide–alkyne cycloaddition (CuAAC) leading to 1,4-substituted triazoles.^[17] This reaction is widely used in organic synthesis, drug discovery, biochemistry, and polymer chemistry.^[18] Now, we have found that the robust zinc reagents of type **1** undergo copper-catalyzed 1,3-dipolar cycloadditions^[19] with in situ generated benzyl azide (**13**) (obtained from NaN₃ and benzyl bromide) or aryl azides **14a,b** with complete regioselectivity.^[20] Smooth deuteration, allylation, amination,^[21] or arylation proved the integrity of the carbon–zinc bond in intermediate **15** and produced the valuable 1,2,3-triazoles **16a–d** in 69–91 % yield (Table 2).^[22] To the best of our knowledge, copper-catalyzed 1,3-dipolar cycloadditions with alkynylzinc derivatives have not been reported previously.^[23]

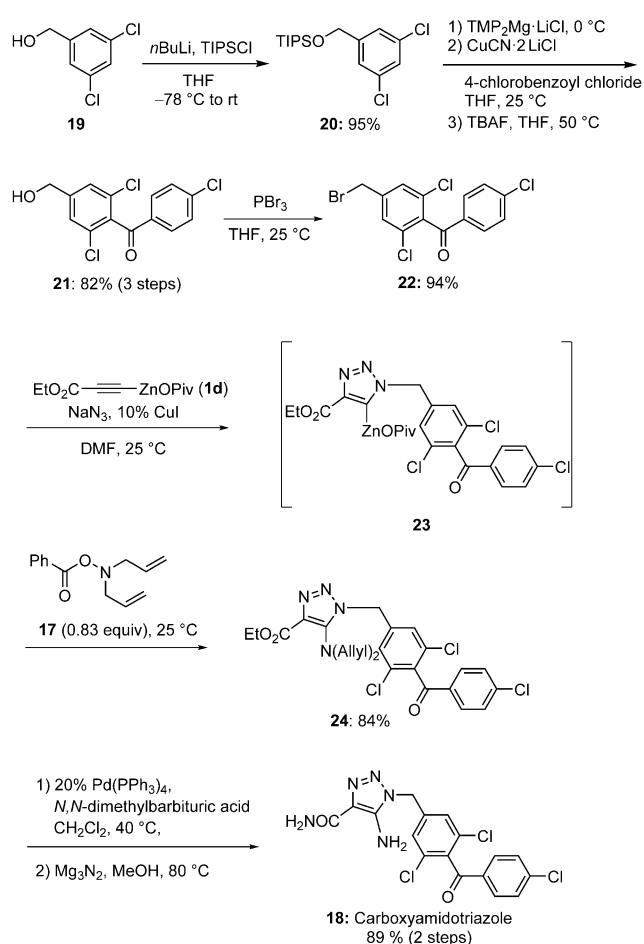
The synthetic utility of alkynylzinc pivalates is further demonstrated in the preparation of an orally active agent carboxyamidotriazole (**18**) with potential antineoplastic activ-

Table 2: Copper-catalyzed dipolar cycloadditions with alkynylzinc pivalates of type **1** providing 1,2,3-triazoles of type **16** via zinc intermediates **15**.

Entry	1	Electrophile	Product
		benzyl azide (13) or aryl azide (14a, b) CuI (10 mol%), DMF 25 °C, 7–21 h	$\text{FG-R} \text{---} \text{C} \equiv \text{C} \text{---} \text{ZnOPiv} \xrightarrow{\text{E-X}} \text{FG-R} \text{---} \text{C} \equiv \text{C} \text{---} \text{E}$ 15 ($\text{R}^2 = \text{benzyl or aryl}$) 16 ($\text{R}^2 = \text{benzyl or aryl}$)
1	1f	$\text{CH}_3\text{CO}_2\text{D}$	16a : 91 %
2	1g	9	16b : 87 %
3	1h	17	16c : 69 %
4	1c	5j	16d : 72 %

ity (Scheme 3).^[24] Thus, the protection of commercially available benzylic alcohol **19** with TIPSCl produced the silyl ether **20** which was smoothly magnesiated at position 4 with $\text{TMP}_2\text{Mg} \cdot \text{LiCl}$ ^[25] followed by copper-mediated acylation^[15] with 4-chlorobenzoyl chloride. After desilylation using TBAF (tetra-*n*-butylammonium fluoride), benzylic alcohol (**21**) was isolated in 82 % yield. The resulting alcohol was brominated with PBr_3 at 25 °C to afford benzylic bromide **22** in 94 % yield. Compound **22** was treated with NaN_3 and the ester-substituted alkynylzinc pivalate **1d** in the presence of 10 % CuI in DMF at 25 °C for 18 h affording the 1,2,3-triazolyzinc pivalate **23**, which was directly aminated with benzoyloxydiallylamine (**17**)^[21] furnishing the trisubstituted 1,2,3-triazole **24** in 84 % yield (based on the amount of **17**) via a four-component one-pot synthesis.^[26] After a Pd-catalyzed deallylation^[27] and an amidation using Mg_3N_2 as reported by Ley,^[28] the desired carboxyamidotriazole (**18**) was obtained in 89 % yield (eight steps, 38 % overall yield).

In summary, we have reported the preparation of a new class of solid alkynylzinc reagents which display enhanced air and moisture stability compared to standard alkynylzinc halides. We have also demonstrated their utility for arylations, allylations, and acylations. Finally, we have shown that the robust zinc reagents **1** undergo 1,3-dipolar cycloadditions without the cleavage of the C–Zn bond which could thus be functionalized in a further step. An efficient preparation of the potential antineoplastic agent **18** demonstrates the utility

**Scheme 3.** Synthesis of carboxyamidotriazole **18** using alkynylzinc pivalate **1d**.

of the new zinc reagents **1**. Further applications are currently underway in our laboratories.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: alkynes · alkynylzinc pivalates · carboxyamidotriazoles · click chemistry · zinc

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[1] Recent reviews: a) *Modern Alkyne Chemistry* (Eds.: B. M. Trost, C.-J. Li), Wiley-VCH, Weinheim, **2014**; b) A. Fürstner, *Angew.*

- Chem. Int. Ed.* **2013**, *52*, 2794–2819; *Angew. Chem.* **2013**, *125*, 2860–2887; c) C. Torborg, M. Beller, *Adv. Synth. Catal.* **2009**, *351*, 3027–3043.
- [2] B. M. Trost, J. T. Masters, *Chem. Soc. Rev.* **2016**, *45*, 2212–2238.
- [3] a) B. M. Trost, J. T. Masters, F. L. Le Vaillant, J.-P. Lumb, *J. Org. Chem.* **2016**, *81*, 10023–10028; b) B. Shrestha, S. Thapa, S. K. Gurung, R. A. S. Pike, R. Giri, *J. Org. Chem.* **2016**, *81*, 787–802; c) S. Tang, L. Zeng, Y. Liu, A. Lei, *Angew. Chem. Int. Ed.* **2015**, *54*, 15850–15853; *Angew. Chem.* **2015**, *127*, 16076–16079; d) R. Kumar, S. Turcaud, L. Micouin, *Org. Lett.* **2014**, *16*, 6192–6195; e) S. Woodward, S. Dagorne, *Modern Organoaluminum Reagents*, Springer, Berlin, Heidelberg, **2013**, p. 312; f) G. A. Molander, B. W. Katona, F. Machrouhi, *J. Org. Chem.* **2002**, *67*, 8416–8423; g) M. Sonoda, A. Inaba, K. Itahashi, Y. Tobe, *Org. Lett.* **2001**, *3*, 2419–2421; h) T. Ooi, T. Miura, K. Takaya, H. Ichikawa, K. Maruoka, *Tetrahedron* **2001**, *57*, 867–873; i) D. E. Frantz, R. Fässler, C. S. Tomooka, E. M. Carreira, *Acc. Chem. Res.* **2000**, *33*, 373–381; j) D. Tzalis, P. Knochel, *Angew. Chem. Int. Ed.* **1999**, *38*, 1463–1465; *Angew. Chem.* **1999**, *111*, 1547–1549.
- [4] a) S. Bernhardt, G. Manolikakes, T. Kunz, P. Knochel, *Angew. Chem. Int. Ed.* **2011**, *50*, 9205–9209; *Angew. Chem.* **2011**, *123*, 9372–9376; b) C. I. Stathakis, S. Bernhardt, V. Quint, P. Knochel, *Angew. Chem. Int. Ed.* **2012**, *51*, 9428–9432; *Angew. Chem.* **2012**, *124*, 9563–9567; c) C. I. Stathakis, S. M. Manolikakes, P. Knochel, *Org. Lett.* **2013**, *15*, 1302–1305; d) J. R. Colombe, S. Bernhardt, C. Stathakis, S. L. Buchwald, P. Knochel, *Org. Lett.* **2013**, *15*, 5754–5757; e) S. M. Manolikakes, M. Ellwart, C. I. Stathakis, P. Knochel, *Chem. Eur. J.* **2014**, *20*, 12289–12297; f) M. Ellwart, P. Knochel, *Angew. Chem. Int. Ed.* **2015**, *54*, 10662–10665; *Angew. Chem.* **2015**, *127*, 10808–10811.
- [5] T. J. Greshock, K. P. Moore, R. T. McClain, A. Bellomo, C. K. Chung, S. D. Dreher, P. S. Kutchukian, Z. Peng, I. W. Davies, P. Vachal, M. Ellwart, S. M. Manolikakes, P. Knochel, P. G. Nantermet, *Angew. Chem. Int. Ed.* **2016**, *55*, 13714–13718; *Angew. Chem.* **2016**, *128*, 13918–13922.
- [6] Y.-H. Chen, M. Ellwart, G. Toupalas, Y. Ebe, P. Knochel, *Angew. Chem. Int. Ed.* **2017**, *56*, 4612–4616; *Angew. Chem.* **2017**, *129*, 4683–4687.
- [7] a) J. M. Hammann, F. H. Lutter, D. Haas, P. Knochel, *Angew. Chem. Int. Ed.* **2017**, *56*, 1082–1086; *Angew. Chem.* **2017**, *129*, 1102–1106. Recent reviews about cobalt catalysis: b) M. S. Hofmayer, J. M. Hammann, F. H. Lutter, P. Knochel, *Synthesis* **2017**, DOI: <https://doi.org/10.1055/s-0036-158898>; c) M. Mose-lage, J. Li, L. Ackermann, *ACS Catal.* **2016**, *6*, 498–525; d) S. Güllak, O. Stepanek, J. Malberg, B. R. Rad, M. Kotora, R. Wolf, A. J. von Wangelin, *Chem. Sci.* **2013**, *4*, 776–784; e) G. Cahiez, A. Moyeux, *Chem. Rev.* **2010**, *110*, 1435–1462; f) C. Gosmini, J.-M. Bégouin, A. Moncomble, *Chem. Commun.* **2008**, 3221–3233 and references therein.
- [8] A. Hernán-Gómez, E. Herd, E. Hevia, A. R. Kennedy, P. Knochel, K. Koszinowski, S. M. Manolikakes, R. E. Mulvey, C. Schnegelsberg, *Angew. Chem. Int. Ed.* **2014**, *53*, 2706–2710; *Angew. Chem.* **2014**, *126*, 2744–2748.
- [9] X. Zhang, T. J. Emge, K. C. Hultsch, *Angew. Chem. Int. Ed.* **2012**, *51*, 394–398; *Angew. Chem.* **2012**, *124*, 406–410.
- [10] a) E.-i. Negishi, G. Wang, H. Rao, Z. Xu, *J. Org. Chem.* **2010**, *75*, 3151–3182; b) E.-i. Negishi, L. Anastasia, *Chem. Rev.* **2003**, *103*, 1979–2017; c) A. O. King, N. O. Okukado, E.-i. Negishi, *J. Chem. Soc. Chem. Commun.* **1977**, 683–684.
- [11] a) R. Martin, S. L. Buchwald, *Acc. Chem. Res.* **2008**, *41*, 1461–1473; b) D. S. Surry, S. L. Buchwald, *Chem. Sci.* **2011**, *2*, 27–50.
- [12] For details on catalysts screening, see the Supporting Information.
- [13] S. Frigoli, C. Fuganti, L. Malpezzi, S. Serra, *Org. Process Res. Dev.* **2005**, *9*, 646–650.
- [14] H. Tokuyama, S. Yokoshima, T. Yamashita, T. Fukuyama, *Tetrahedron Lett.* **1998**, *39*, 3189–3192.
- [15] P. Knochel, M. C. P. Yeh, S. C. Berk, J. Talbert, *J. Org. Chem.* **1988**, *53*, 2390–2392.
- [16] J. Shannon, D. Bernier, D. Rawson, S. Woodward, *Chem. Commun.* **2007**, 3945–3947.
- [17] a) V. V. Rostovtsev, L. G. Green, V. V. Fokin, K. B. Sharpless, *Angew. Chem. Int. Ed.* **2002**, *41*, 2596–2599; *Angew. Chem.* **2002**, *114*, 2708–2711; b) C. W. Tornøe, C. Christensen, M. Meldal, *J. Org. Chem.* **2002**, *67*, 3057–3064.
- [18] a) J. E. Hein, V. V. Fokin, *Chem. Soc. Rev.* **2010**, *39*, 1302–1315; b) S. K. Mamidiyala, M. G. Finn, *Chem. Soc. Rev.* **2010**, *39*, 1252–1261; c) M. Meldal, C. W. Tornøe, *Chem. Rev.* **2008**, *108*, 2952–3015.
- [19] R. Huisgen, *Angew. Chem. Int. Ed. Engl.* **1963**, *2*, 565–598; *Angew. Chem.* **1963**, *75*, 604–637.
- [20] The regioselectivity was confirmed by NMR-analysis.
- [21] a) A. M. Berman, J. S. Johnson, *J. Am. Chem. Soc.* **2004**, *126*, 5680–5681; b) M. Campbell, J. S. Johnson, *Org. Lett.* **2007**, *9*, 1521–1524; c) S. L. McDonald, C. E. Hendrick, Q. Wang, *Angew. Chem. Int. Ed.* **2014**, *53*, 4667–4670; *Angew. Chem.* **2014**, *126*, 4755–4758.
- [22] a) F. Wei, T. Zhou, Y. Ma, C.-H. Tung, Z. Xu, *Org. Lett.* **2017**, *19*, 2098–2101; b) W. Wang, X. Peng, F. Wei, C.-H. Tung, Z. Xu, *Angew. Chem. Int. Ed.* **2016**, *55*, 649–653; *Angew. Chem.* **2016**, *128*, 659–663; c) W. Wang, F. Wei, Y. Ma, C.-H. Tung, Z. Xu, *Org. Lett.* **2016**, *18*, 4158–4161; d) F. Wei, W. Wang, Y. Ma, C.-H. Tung, Z. Xu, *Chem. Commun.* **2016**, 52, 14188–14199; e) F. Wei, H. Li, C. Song, Y. Ma, L. Zhou, C.-H. Tung, Z. Xu, *Org. Lett.* **2015**, *17*, 2860–2863; f) B. T. Worrell, S. P. Ellery, V. V. Fokin, *Angew. Chem. Int. Ed.* **2013**, *52*, 13037–13041; *Angew. Chem.* **2013**, *125*, 13275–13279; g) Y. Zhou, T. Lecourt, L. Micouin, *Angew. Chem. Int. Ed.* **2010**, *49*, 2607–2610; *Angew. Chem.* **2010**, *122*, 2661–2664; h) D. V. Partyka, L. Gao, T. S. Teets, J. B. Updegraff III, N. Deligonul, T. G. Gray, *Organometallics* **2009**, *28*, 6171–6182; i) X. Zhang, R. P. Hsung, H. Li, *Chem. Commun.* **2007**, 2420–2422.
- [23] a) A. Krasinski, V. V. Fokin, K. B. Sharpless, *Org. Lett.* **2004**, *6*, 1237–1240; b) A. Akao, T. Tsuritani, S. Kii, K. Sato, N. Nonoyama, T. Mase, N. Yasuda, *Synlett* **2007**, 0031–0036; c) C. D. Smith, M. F. Greaney, *Org. Lett.* **2013**, *15*, 4826–4829; d) Y. Li, X. Qi, Y. Lei, Y. Lan, *RSC Adv.* **2015**, *5*, 49802–49808.
- [24] R. Ju, L. Guo, J. Li, L. Zhu, X. Yu, C. Chen, W. Chen, C. Ye, D. Zhang, *Cancer Lett.* **2016**, *370*, 232–241.
- [25] a) A. Unsinn, C. J. Rohbogner, P. Knochel, *Adv. Synth. Catal.* **2013**, *355*, 1553–1560.
- [26] a) C. Zhou, J. Zhang, P. Liu, J. Xie, B. Dai, *RSC Adv.* **2015**, *5*, 6661–6665; b) Y. Jiang, D. Kong, J. Zhao, W. Zhang, W. Xu, W. Li, G. Xu, *Tetrahedron Lett.* **2014**, *55*, 2410–2414; c) T. Cook, J. A. Walker, Jr., J. Mack, *Green Chem.* **2013**, *15*, 617–619; d) S. Mohammed, A. K. Padala, B. A. Dar, B. Singh, B. Sreedhar, R. A. Vishwakarma, S. B. Bharate, *Tetrahedron* **2012**, *68*, 8156–8162.
- [27] F. Garro-Helion, A. Merzouk, F. Guibé, *J. Org. Chem.* **1993**, *58*, 6109–6113.
- [28] G. E. Veitch, K. L. Bridgwood, S. V. Ley, *Org. Lett.* **2008**, *10*, 3623–3625.

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